Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Ruff, Gary A. Ph.D.		
Project Title:	Combustion of Unsupported Droplet Clusters in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44135-3191	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
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No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Choi, Mun-Young (Drexel University)		
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Combustion experiments using arrays of droplets seek to provide a link betwen single droplet combustion phenomena and the behavior of complex spray combustion systems. Both single droplet and droplet array studies have been conducted in microgravity to better isolate the droplet interaction phenomena and eliminate or reduce the confounding effect of buoyancy-induced convection. In most experiments involving droplet arrays, the droplets are supported on fibers to keep them stationary and close together before the combustion event. The presence of the fiber, however, disturbs the combustion process by introducing a source of heat transfer and asymmetry into the configuration. As the number of drops in a droplet array increases, supporting them on fibers becomes less practical because of the cumulative effect of the fibers on the combustion process. To eliminate the effect of the fiber, several researchers have conducted microgravity experiments using single unsupported droplets -- this is the first experiment using multiple unsupported droplets. The overall objective of this research is to extend the study of unsupported droplets by investigating the combustion of well-characterized drop clusters in microgravity. This will be accomplished by forming and stabilizing a

Task Description:	cluster of droplets in an acoustic field prior to the microgravity and combustion event. The number and size of the drops in the cluster can be accurately controlled using a custom on-demand droplet generator and the size and shape of the cluster can be controlled through the design of the acoustic resonance chamber and selection of the resonance frequency. Additionally, group combustion numbers (ratio of the mass transfer between the phases in the cloud to the mass transfer between the ambiance and the cloud surface) between 0.1 and 100 can be obtained which encompasses the individual particle combustion, group combustion, and sheath combustion regimes. Because the presence of an acoustic field greatly influences drop combustion, the field will be turned off at the start of the microgravity period. The initial diagnostic capability to be incorporated into the experimental package includes multiview imaging to view the motion of the droplets and to view the position of the flame front. Initially, droplet clusters will be formed using methanol, n-decane, and heptane as many of the droplet array and group combustion studies have done. This will allow the droplet burning rates obtained from this research to be directly compared with results of isolated droplet and droplet array studies. The position and movement of the flame front will provide some of the first experimental data for comparison with group combustion models obtained using large numbers of drops. Data will be obtained as a function of group combustion number, drop size, number of drops, and drop size distribution. Several tests are also planned in which drops will be formed with binary fuels composed of heptane/methanol, heptane/ethanol, or heptane/hexadecane to experimentally investigate the effect of fuel volatility on cluster combustion and help define future research. These fuel compositions are the same as those used in other drop combustion research, immediately linking the current results to existing research. The specific objective of this research is to study the combustion of well-characterized drop clusters to better link isolated droplet and droplet array studies with those involving spray flames. In this study, which will be conducted in a series of drop tower experiments, the formation of the drop clusters will be precisely controlled using an acoustic levitation system so that dilute, as well as dense clusters can be created and stabilized before combustion is begun. Tasks to be accomplished during this research include (1) the design and fabrication of a unique droplet combustion experiment using acoustic levitation to stabilize a droplet cluster prior to combustion, (2) investigation of droplet combustion in the individual particle combustion, group combustion, and sheath combustion regimes by varying the number of droplets, droplet size, and cluster diameter, and (3) investigation of the effect of drop size distribution, droplet composition, and continuous-phase oxygen concentration on cluster combustion. Isolated drop and drop array studies have become common methods to isolate and investigate the many complexities of the drop combustion process. Accordingly, several isolated drop, drop array, and spray studies are currently in progress under the NASA Microgravity Science and Applications program. Results from these research programs indicate that even with the effects of buoyancy removed, the extrapolation of the results from droplet array studies to spray flames is difficult. The problem arises because even the simplest spray systems introduce complexities of multi-disperse drop sizes and drop-drop interactions, coupled with more complicated fluid dynamics. Not only do these features make the interpretation of experimental data difficult, they make the problem very difficult to analyze computationally. Group combustion models have become a popular method to investigate the behavior of large numbers of interacting droplets. In these models, the interaction between droplets is treated in a statistical manner rather than tracking and evaluating drop interactions using a Lagrangian approach. These models have allowed very comprehensive investigations of the combustion of multi-component drops, spray ignition phenomena, high-pressure combustion, etc. While these models idealize actual spray systems to a point where they can be treated computationally, the experimental analogy to these models is difficult to achieve. Therefore, even though these models have provided useful results, the verification of the results with direct comparison with experimental data is still lacking. Direct experimental measurements in well-defined droplet clusters would fill a large gap in our current understanding of droplet combustion. The results of this study will provide an intermediate link between droplet array studies and those performed using full combusting spray systems. They will also provide unique experimental data for the calibration, verification, and improvement of group combustion models that are currently being applied to complex flow situations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)